Bidirectional Selective State-Space Models for Food-Texture Classification from Chewing-Behavior Video

A sequence-model study on a 20-subject cohort

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Motivation: why texture from chewing video?

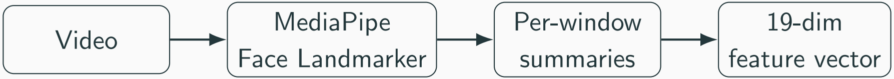
- Clinical stake: dietary monitoring, nutrition assessment, and assistive-feeding systems benefit from non-intrusive, camera-only measurement of chewing behavior.
- Hard because it is *temporal*: texture modulates jaw-cycle frequency, slow mouth-opening drift, and pause structure — properties of a sequence, not of a single snapshot.

Problem statement: the classification task

- 8,077 windows $\times$ 19 landmark-derived features $\times$ 20 subjects.
- Expert-annotated labels over three texture classes: soft, medium, hard.
- Objective: per-window classification, generalizing across unseen subjects.

Feature pipeline: video to 19-dim feature vector

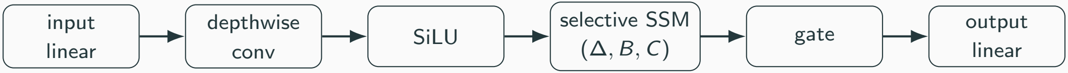
Per-window summaries of mouth-open amplitude, mouth width, jaw y-coordinates, smoothed mouth opening, jaw lateral delta, pause ratio, and window duration.



Model family: tabular baselines vs. sequence model

- **Classical:** Logistic Reg. / RF / XGBoost on per-window feature vectors.
- **Tabular NN:** MLP ($128 \to 64 \to 32$), FT-Transformer.
- **Sequence model (ours):** Bidirectional Mamba-1 via `mamba-ssm` fused CUDA kernel; $d_{\text{model}} = 64$, $n_{\text{blocks}} = 2$, $d_{\text{state}} = 16$.

Inside one Mamba block



Bidirectional: forward and reversed Mamba layers are summed inside each residual block.

Cross-validation: 5-fold StratifiedGroupKFold by subject

- Group key: `video_id` (subject); 5 folds $\times$ 4 held-out subjects each.
- Leave-subjects-out: no subject appears in both train and validation within a fold.
- Metrics: per-window macro-F1 (primary); per-chunk majority-vote macro-F1 (secondary).
- Feature means/stds and class weights computed on the training fold only.

Main result — at-a-glance

- Mamba achieves window macro-F1 $= \mathbf{0.630} \pm \mathbf{0.098}$ at the $\sigma = 0.1$ reference regime.

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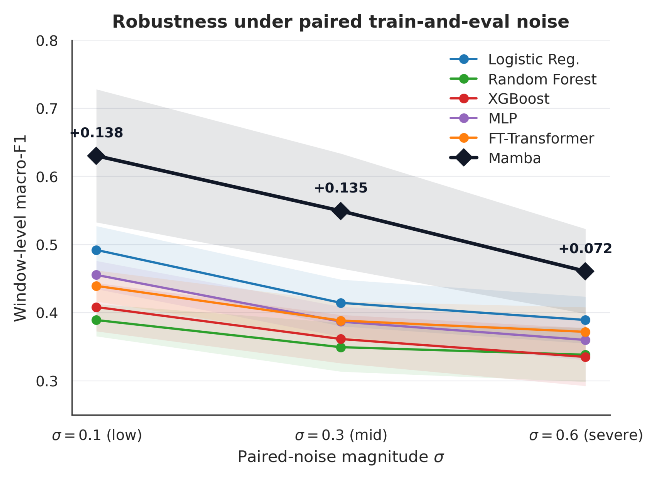
- Mamba achieves window macro-F1 $= \mathbf{0.630} \pm \mathbf{0.098}$ at the $\sigma = 0.1$ reference regime.
- Best tabular baseline (logreg) lands at 0.492: a $+\mathbf{0.138}$ window-F1 lead for the sequence model.

Main result — full table

Model	Window macro-F1	Chunk macro-F1
logreg	0.492 ± 0.035	**0.685**
rf	0.389 ± 0.024	0.430
xgb	0.408 ± 0.035	0.440
mlp	0.455 ± 0.021	0.526
ftt	0.439 ± 0.023	0.452
Mamba-1 (ours)	$\mathbf{0.630 \pm 0.098}$	0.652

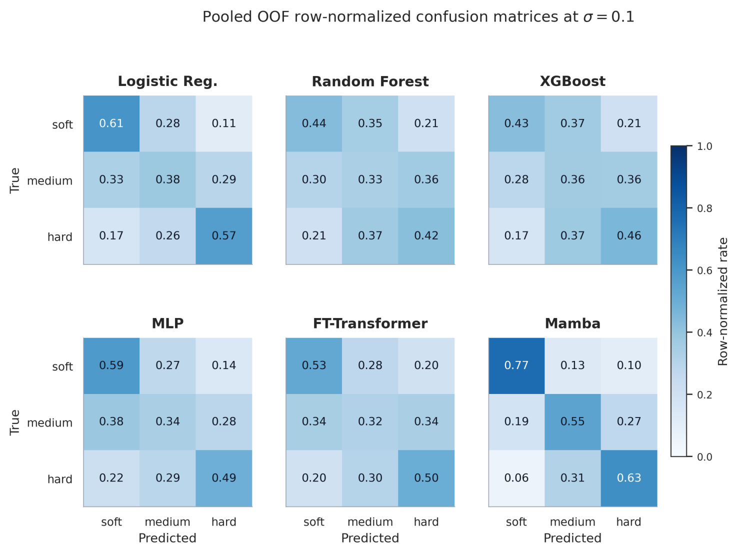
Fold-mean $\pm$ std at the $\sigma = 0.1$ reference regime. Caveat: logreg narrowly wins chunk-F1 (0.685 vs 0.652) because chunk-consistent per-window summaries benefit majority-vote aggregation.

Robustness: paired-noise train-and-eval regime



Per-window macro-F1 under a paired-noise train-and-eval regime at $\sigma \in \{0.1, 0.3, 0.6\}$. Mamba retains its absolute lead across all magnitudes; the gap narrows at $\sigma = 0.6$ as all models converge

Per-class behavior: medium is the hardest class



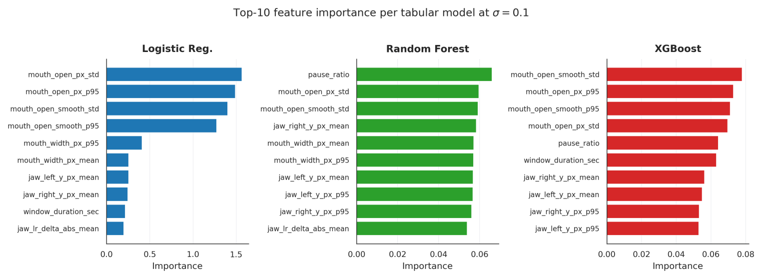
Ablation: Mamba-3 (hybrid) vs. Mamba-1 (matched-size baseline)

- Mamba-3 (hybrid) wins by $+0.023$ window macro-F1 and $+0.046$ chunk macro-F1.
- Wall-clock cost $\approx 143\times$ the fused kernel (implementation-driven, not algorithmic).

Variant	Window macro-F1	Chunk macro-F1
Mamba-1 (matched-size baseline)	0.373 ± 0.036	0.446
Mamba-3 (matched-size hybrid)	0.396 ± 0.026	0.492
Δ	$+0.023$	$+0.046$

Matched-size config: $d_{\text{model}} = 32$, $n_{\text{blocks}} = 1$, $d_{\text{state}} = 8$, 30 epochs — not comparable to the main Mamba result on Slide 9.

Feature importance: what the tabular models agree on



Mouth-open variability (std, p95) is the dominant signal across all three tabular models; pause_ratio is additionally highly ranked by rf and xgb. Harder foods elicit larger-amplitude, more variable chewing motion with fewer low-activity intervals.

Limitations and next steps

- Cohort size: 20 subjects, 4 per held-out fold — fold-mean numbers carry real uncertainty.
- Per-fold Mamba window-F1 std $= 0.098$, largest among models: subject-level instability.
- Ablation trained at 30 epochs (tiny config) vs 80 epochs (main) — budget-matched comparison is future work.
- Single seed per fold; multi-seed averaging would harden the ranking.
- Follow-ups: larger cohort; GRU / TCN / small-Transformer sequence baselines; fused CUDA kernel for Mamba-3 upgrades; finer-grained (frame-level) modeling; multi-modal fusion.

Conclusion and Q&A

- Selective state-space models are a credible inductive-bias choice for chewing-video texture classification.
- Headline: $+0.138$ window-F1 over the strongest tabular baseline at $\sigma = 0.1$; preserves a $+0.072$ lead even at $\sigma = 0.6$.
- Open: specifically *selective* SSMs vs. other sequence families; scaling to larger cohorts; fused kernels for Mamba-3-style refinements.

Questions?